Alma Models

Independent Audit, Improvements and Final Outcome

Full report of the work done on 1 November 2026

Item	Detail
Repository	github.com/Bhavith-Chandra/Alma-Models (audited at commit 731edfa)
Date of work	1 November 2026 (single working session, continued over several restarts)
Prepared by	Claude Code (Opus 5.5), working on the user's machine
Scope	Test every model with its own harness, look for data leaks, faked numbers and over-perfect testing; improve the models with small experiments and scale up what works
Where the work is	Local folder *Alma Models/*: *repo/* (branch *audit-fixes*, 2 commits) and *audit/* (all audit scripts, logs and REPORT.md). Nothing pushed to GitHub.

1. Executive summary (final outcome)

Final outcome in one paragraph

- **No data leaks and no faked numbers** were found in the four headline models (oil, gas, power, crypto volatility). Every published TEST score was re-produced from freshly downloaded public data, with independent scoring code, and every model passed a leak test.
- **Some claims were misleading** (cherry-picked comparisons): the crypto "transfer to unseen coins" claim, the oil "ranks #1 of 9" claim, and the "world model" idea itself, including the research-track SSM that was said to be "the first to pass the world-model test". All of these have been corrected in the docs.
- **Three models were improved** and packaged: gas (MAE 10.6 vs 11.5 Bcf, better and narrower intervals), power (quantile score 8.60 vs 9.03) and crypto volatility (direction 74.0% vs 72.8%).
- A **forward test** was frozen so the improved models can be judged on data that did not exist when they were built. Cross-machine **reproducibility** was added.
- Everything is committed on the local branch *audit-fixes*; **pushing to GitHub is waiting for your approval**.

Scorecard

Question	Answer
Are there data leaks?	**No.** Scrambling all future data changed 0 forecasts in every project.
Are any numbers faked?	**No.** Oil, gas, power and crypto TEST scores reproduce (exactly for oil and gas).
Is the testing "too perfect"?	The testing itself is sound (pre-registered TEST year, sha256 pins). The **framing** of some results is too generous; corrected.
Does the "world model" idea work?	**No measurable benefit anywhere.** A matched model without the rollout is equal or better in all four projects and in the research SSM.
Could the models be improved?	**Yes** for gas, power and crypto, with ensembles and blends chosen only on development data. Oil is a random walk; nothing to add.
Is everything finished?	Yes for all work possible today. Waiting only on (1) your push approval and (2) new data for the forward test (first gas data 2026-11-05).

2. What was asked and how it was tested

The request: go through the Alma Models repository, test the models with their own harness "with no faking", find models with suspiciously perfect testing, data leaks or number faking, try to improve them with small experiments, and scale up the ones that do better.

Testing method (applied to every project)

- **Rebuild the data from the public sources** (FRED, EIA, NOAA CPC, Energy-Charts, Binance) and compare every file to the sha256 pins stored in the repo. A model can only be trusted if it is scored on the same data.
- **Run the frozen checkpoints through their own wrappers**, i.e. the exact files the repo says were scored.
- **Score with my own metric code**, written independently of the repo's scorers (pinball/quantile score, MAE, direction accuracy, interval coverage).
- **Leak test:** scramble every value the model must not know at forecast time. If any forecast changes, the model is peeking at the future.

- **Genuineness check:** retrain from the stored recipe and compare with the frozen checkpoint, to confirm the weights really came from the stated recipe and training period.
- **Significance:** block bootstrap (by day or week) and family-cluster bootstrap confidence intervals, so a "win" is only called a win when it is bigger than the noise.
- **Discipline for improvements:** every choice was made on DEV/VAL data only, frozen in writing, and only then scored once on TEST.

3. Verification results: leaks and faked numbers

Project	Data rebuilt vs sha256 pins	Checkpoint genuine?	Leak test	Published TEST numbers reproduced?
Oil (wm_sv)	FRED files byte-identical	Yes: retrained from recipe, weights match to 7e-7	0/25 changed	**Exactly** (QS 0.01052 / 0.02383 / 0.04994)
Gas (wm_gas)	EIA storage + CPC forecast archive byte-identical	Yes: refit seed 0, outputs corr 0.99996	0/26 changed	**Exactly** (MAE 11.53/11.56, acc 0.788/0.808)
Power (wm_power)	Energy-Charts 2015-2025 byte-identical (2026 file 1 day longer)	Wrapper sha-checked	0/25 changed	Yes (P2 0.746 vs 0.747; P1 0.731 vs 0.732; 1-day window difference)
Crypto vol (wm_fin)	All 4,700 Binance files + universe + sealed TEST byte-identical	DEV predictions re-run: corr 0.99997	Harness asserts	Yes (P1 0.727/0.728; QS 0.1459/0.1496)

Verdict

- No data leakage and no fabricated numbers in the four headline models. The checkpoints are real, trained only on pre-test data, and their TEST scores reproduce.

4. What was wrong: misleading claims

The numbers are honest, but several headline claims were framed to look stronger than the evidence supports.

#	Claim in the repo	What the evidence shows	Status
1	Crypto: "transfers to unseen coins, +8-9 pp, Holm < 1e-4"	Only measured against two baselines that have no weekday input. The repo's own weekday-aware HAR (har_intra_dow) scores **0.731 vs wm_fin 0.728** on held-out coins: no advantage.	Corrected
2	Oil: "wm_sv ranks #1 of 9"	Skill vs random walk +0.45%, p = 0.48 (noise). A simple EWMA(0.94) random walk scores 0.01054 vs 0.01052 at 1 day and beats it at 5 days. wm_sv is effectively a random walk.	Corrected
3	The "world model" (latent rollout) is what makes the models good	The matched direct twin (no rollout) is equal or better: gas TEST MAE 11.40 vs 11.53, crypto QS 0.1465 vs 0.1471, oil and power similar.	Corrected
4	Power: "#1 numerically on hourly direction"	0.747 vs Chronos-2 0.740 is not significant (the repo does disclose this).	Already disclosed
5	Research SSM: "the first to pass the world-model test"	Based on seed 0 only, which is also the seed that selected the SSM. Over 5 seeds it does not pass (see section 6.6).	Corrected
6	Older research leaderboard (38 numbers)	The repo's own audit_v3 says none of these is traceable to code + config. Nothing to reproduce from.	Noted

5. Improvement experiments

Approach: start small on the development backtest, scale up only what helps, freeze the choice, then look at TEST once. **Caveat:** the TEST year had already been scored by the repo, so these TEST numbers are a *second look*, not a pristine hold-out. That is why a forward test was also set up (section 6.4).

5.1 Gas storage (biggest win)

- **Small experiment** (6 rolling DEV folds, 2019-07 to 2025-06): pool rollout + direct seeds and blend with the OLS baseline, weight chosen leave-one-fold-out. DEV MAE 16.59 to 15.72; better in 5 of 6 folds; fold weights stable at 0.2-0.3.
- **Scaled up:** trained 8 new DEV backtest runs (seeds 3-6 of both recipes) giving a 14-member pool + 0.20 OLS. DEV MAE 15.39 vs 16.59 for the frozen recipe (-7%). Frozen in GAS_FREEZE.md, then refit seeds 3-6 on all training weeks.
- **Interval fix** (amendment 1, chosen on DEV): interval width now scales with the recent 8-week error level, (recent |error| / DEV |error|)^0.5, clipped to 0.4-2.0.

Model (TEST, Lower 48, 52 weeks)	MAE thu / fri (Bcf)	Quantile score thu / fri	80% coverage	Direction thu / fri
wm_gas (original, frozen)	11.53 / 11.56	4.59 / 4.56	-	0.788 / 0.808
st_ols_wx (OLS baseline)	14.35 / 14.46	6.14 / 6.09	-	0.808 / 0.808
14-seed pool + 0.2 OLS	10.63 / 10.46	4.30 / 4.23	0.94 (too wide)	0.808 / 0.788
wm_gas_ens (pool + adaptive intervals)	**10.63 / 10.46**	**4.14 / 4.07**	0.885 / 0.904	0.808 / 0.788

- vs OLS: MAE -3.7 / -4.0 Bcf, confidence interval excludes 0 (the original model was **not** significant vs OLS).
- vs original wm_gas: MAE -0.9 / -1.1 (p 0.13 / 0.08): better but not significant on 52 weeks.
- Adaptive intervals: quantile score -0.16 (CI excludes 0), intervals 17% narrower. Leak test: 0/22 changed.

5.2 Crypto volatility

- Quantile blend 0.6 x wm_fin + 0.4 x har_intra_dow; weight chosen on VAL (2025-07 to 2025-10) only.
- Result is stable for weights 0.3 to 0.5 (all reported, not just the best).

Model (TEST, 24 coins)	Direction accuracy	Quantile score
wm_fin (original)	0.724 (median rule) / 0.728 (published)	0.1471
har_intra_dow (baseline)	0.731	-
wm_fin_blend	**0.740**	**0.1459**

vs wm_fin: +1.7 pp, CI [+0.5, +2.9] (significant). vs har_intra_dow: +0.9 pp, CI [-0.6, +2.3] (not significant on direction), but quantile score -0.008 with CI excluding 0.

5.3 German day-ahead power

- Quantile blend 0.5 x wm_power + 0.5 x LEAR; weight chosen on VAL.

Model (TEST)	Quantile score	MAE	Hourly direction (P2)
wm_power (original)	9.03	22.4	0.746
LEAR (baseline)	9.22	-	-
wm_power_blend	**8.60**	**21.5**	**0.755**

Quantile score is significantly better than both members. Direction is not significant vs wm_power, significant vs LEAR.

5.4 Oil

No improvement attempted. The model is indistinguishable from a random walk, so there is no signal for an ensemble or blend to add. Claiming an improvement here would itself be number-faking.

6. Follow-up work (all completed)

6.1 Claims corrected

README.md, winning_models/README.md and the oil and crypto model cards now say: oil ties the random walk; the crypto transfer claim names the weekday HAR that matches it on held-out coins; the direct twin is equal or better everywhere; and they point to this audit. The research-program doc carries a correction for the SSM claim.

6.2 Gas intervals fixed

Cause: interval width was nearly constant (about 52 Bcf) whatever the recent error level. New rule chosen leave-one-fold-out on DEV: DEV QS 5.97 to 5.92; TEST QS 4.30 to **4.14** (significant), coverage 0.94 to 0.89/0.90, intervals 17% narrower.

6.3 Improved models packaged

Model	Wrapper	Folder	Check
wm_gas_ens	tools/gas/wm_gas_ens.py	winning_models/gas_wm_gas_ens/	14 sha-checked members; harness reproduces offline numbers exactly; leak test 0/22
wm_fin_blend	tools/fin/wm_fin_blend.py	winning_models/fin_wm_fin_blend/	Harness reproduces offline numbers exactly
wm_power_blend	tools/power/wm_power_blend.py	winning_models/power_wm_power_blend/	Harness reproduces offline numbers exactly

6.4 Forward test (true out-of-sample check)

- tools/forward/FREEZE.json pins 29 code and checkpoint files (frozen 2026-11-01 17:47 UTC).
- forward_eval.py refresh downloads new public data; score evaluates only periods after 2026-11-01 and **refuses to run if any pinned file has changed**.
- Dry run on old data works for all three projects (written to a folder clearly named DRYRUN_NOT_A_RESULT).
- First gas data: Thursday 2026-11-05. A meaningful result needs about 20 gas weeks or about 60 days for power and crypto.

6.5 Reproducibility

Training is already bit-identical run-to-run on one machine. The small drift vs the original checkpoints comes from different hardware (multithreaded sums). A new ALMA_THREADS=1 switch in all trainers gives single-threaded, deterministic training across machines (oil reproduced across machines to 7e-7). Default behaviour is unchanged (verified bit-identical). Documented in docs/reproducibility.md.

6.6 Research track: the SSM world-model test, settled over 5 seeds

Published checkpoints re-scored on the bench VAL split (30 tasks): rollout 0.6706 (claimed 0.6707), direct 0.6926 (claimed 0.6926): exact. The direct twins for seeds 1-4 had never been trained, so I trained the 8 missing runs with the exact stored recipe (lower score is better):

Seed	Rollout (published model)	Published claim	Direct twin (no rollout)	Rollout better?
0 (selection seed)	0.6706	0.6707	0.6926	Yes
1	0.6947	0.6936	0.6789	No
2	0.6804	0.6804	0.6746	No
3	0.6808	0.6808	0.6951	Yes
4	0.6765	0.6765	0.6861	Yes

Rollout better on 3 of 5 seeds. Mean gain +0.7%, 95% CI [-0.03%, +1.3%]: **not significant**. Without the selection seed: +0.08%, effectively zero. **Verdict: the SSM does not pass the world-model test.** Written up in docs/ssm_wm_test_5seed.md. (Seed 1 differs by 0.001 because its run was interrupted and resumed.)

7. All changes made

7.1 Git commits (local branch audit-fixes, not pushed)

Commit	Message	Summary
9f96b36	Audit follow-up: correct claims, add ensemble/blend models, forward test	Claim corrections, 3 new models + cards + checkpoints, forward test, reproducibility switch
7f8c068	Research track: 5-seed SSM world-model test (not passed); correct the seed-0 claim	New 5-seed write-up and correction in the research-program doc

Total: 36 files changed, 699 lines added, 13 changed, on top of the original commit 731edfa.

7.2 Files changed in the repository

Area	Files	What changed
Docs and claims	README.md; winning_models/README.md; winning_models/oil_wm_sv/MODEL_CARD.md; winning_models/fin_wm_fin/MODEL_CARD.md; docs/almats_research_program.md	Misleading claims corrected; follow-up models table added
New docs	docs/reproducibility.md; docs/ssm_wm_test_5seed.md	Cross-machine drift and ALMA_THREADS rule; 5-seed SSM result
New gas model	tools/gas/wm_gas_ens.py; winning_models/gas_wm_gas_ens/ (MODEL_CARD.md, 2 checkpoints, SHA256SUMS); 9 prediction files + 9 rows in runs/gas/search/	14-seed ensemble + OLS blend + adaptive intervals
New crypto model	tools/fin/wm_fin_blend.py; winning_models/fin_wm_fin_blend/MODEL_CARD.md	0.6 wm_fin + 0.4 har_intra_dow
New power model	tools/power/wm_power_blend.py; winning_models/power_wm_power_blend/MODEL_CARD.md	0.5 wm_power + 0.5 LEAR
Forward test	tools/forward/forward_eval.py, FREEZE.json, README.md	Freeze / refresh / score with a frozen-hash guard
Trainers	tools/gas/search/run.py; tools/power/search/run.py; tools/fin/search/{loop, train_wm, finalize, train_agents}.py	ALMA_THREADS switch for deterministic training (default unchanged)
Results log	runs/commodity/search/results.tsv	One reproduction row for oil

7.3 Audit folder (outside the repo)

Purpose	Scripts
Independent re-scores + leak tests	oil_check.py, gas_check.py, power_check.py, fin_check.py, fin_score.py
Gas experiments	gas_exp.py, run_gas_jobs.py, gas_final.py, gas_calib.py, gas_final2.py, gas_ens_leak.py, GAS_FREEZE.md
Crypto and power	fin_val.py, fin_blend.py, fin_boot.py, power_blend.py, power_wrap_check.py
Research track	alma_bench_check.py, run_ssm.sh, ssm_lanes.sh, ssm_analysis.py
Data	fetch_fin_from_manifest.py
Report	REPORT.md (full technical write-up)

Not committed on purpose: rebuilt raw data (e.g. data/bench/) and the audit seed checkpoints in data/ckpt/. They are large and can be rebuilt from the scripts.

8. Problems met and how they were solved

Problem	Fix
Project folder name ends with a space	Quoted every path
NOAA CPC forecast files missing from the clone	Downloaded all 978 from the original manifest URLs; restored manifest_cpc.json; result byte-identical to the pins
universe_candidates.txt missing	Rebuilt from the klines_1d listing; universe.json sha matched
Crypto TEST loader is sealed	Opened with ALMA_FIN_TEST_GO=1: legitimate because the TEST year had already been scored; this is an audit re-score
Gas and crypto both use a module named data	Run one subprocess per project
setsid not available on macOS	Removed it
Background training killed by session restarts	Moved runs to a terminal tab; trainer resumed seed 1 from its saved checkpoint (steps 1554 / 1495)
SSM log parser failed	Fixed the header detection regex
One status message mislabelled seed-1 numbers	Corrected in the next status update

9. Final outcome and what remains

Final outcome

- **Integrity:** the repo is honest. No leaks, no faked numbers; every headline TEST score reproduces.
- **Claims:** five over-generous claims found and corrected (crypto transfer, oil #1, world-model benefit, research SSM, plus the untraceable old leaderboard noted).
- **Main scientific finding:** the "world model" rollout adds nothing measurable anywhere. The gains come from good features, seed ensembles and blending with the strongest simple baseline.
- **Improvements delivered:** gas MAE -8 to -10% and better intervals; power quantile score -5%; crypto +1.7 pp direction over wm_fin. None beats its best simple baseline significantly on direction; gas now significantly beats OLS on MAE.
- **Future-proofing:** frozen forward test and deterministic training switch.

What remains

Item	Status	Who / when
Push branch audit-fixes to GitHub and open a pull request	Ready	Waiting for your approval
Forward test results	Built and frozen	First gas data 2026-11-05; meaningful after ~20 weeks (gas) or ~60 days (power, crypto)
Optional: weekly automatic forward-test run	Not set up	Can be added on request

Appendix: glossary

Term	Meaning
TEST year	The pre-registered evaluation period (2025-11 to 2026-10) that models must never see in training
DEV / VAL	Earlier periods used to make every choice before TEST is looked at
Leak test	Scramble data from the future; a clean model's forecast must not change
MAE	Mean absolute error (lower is better)
Quantile score (QS)	Pinball loss over forecast quantiles; scores the whole forecast range (lower is better)
Coverage (80%)	Share of outcomes inside the 80% interval; ideal is 0.80
Direction accuracy	How often the model gets up vs down right
World model / rollout	A model that simulates future latent states step by step
Direct twin	The same model trained to predict directly, with no rollout, used as a fair comparison
Bootstrap CI	Resampling-based confidence interval; a win counts only if the interval excludes 0
sha256 pin	A fingerprint proving a file is byte-for-byte the one that was tested
Forward test	Scoring frozen models only on data published after they were frozen